

Advanced approximate and symbolic computing

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Function interpolation and approximation III.

Approximation in normed linear space

Best approximations (in normed linear spaces)

- Given a normed vector space $\mathcal{V}$ and its subspace $\mathcal{G}$ and an arbitrary element $\mathbf{f} \in \mathcal{V}$.
- If $\mathbf{f} \notin \mathcal{G}$, then $\mathbf{f}$ cannot be given with the elements of $\mathcal{G}$.
- In this case we need to have the linear combination

$$\mathbf{f} \approx \mathbf{p} = \sum_{i=1}^k \alpha_i \mathbf{g}_i.$$

- The error of the approximation $\mathbf{f} \approx \mathbf{p}$ is defined as $\|\mathbf{f} - \mathbf{p}\|$.
- We can ask: which element $\mathbf{p}^*$ gives the *best approximation*,
- or put it differently, for which element $\mathbf{p}^* \in \mathcal{G}$ gives the minimal error.

Best approximation in the subspace $\mathcal{G} \subseteq \mathcal{V}$ altern

Fix a basis $\{\mathbf{g}_1, \mathbf{g}_2, \dots, \mathbf{g}_n\}$ of $\mathcal{G}$, meaning that

$\mathcal{G} = \text{span}(\mathbf{g}_1, \mathbf{g}_2, \dots, \mathbf{g}_n)$, and consider the following problem.

Given the approximations of the form $\mathbf{p} = \sum_{i=1}^n \alpha_i \mathbf{g}_i$ of the element $\mathbf{f} \in \mathcal{V}$ find the *best approximation* $\mathbf{p}^*$ for which

$$\|\mathbf{f} - \mathbf{p}^*\| \leq \|\mathbf{f} - \mathbf{p}\| \quad (1)$$

for any element $\mathbf{p} \in \mathcal{G}$.

- For an arbitrary $\mathbf{f} \in \mathcal{V}$ let $E_{\mathcal{G}}(f) \stackrel{\text{def}}{=} \inf\{\|\mathbf{f} - \mathbf{p}\| : \mathbf{p} \in \mathcal{G}\}$.
- Hence an element $\mathbf{p}^*$ is the best approximation of $\mathbf{f}$ in $\mathcal{G}$ if and only if $\|\mathbf{f} - \mathbf{p}^*\| = E_{\mathcal{G}}(f)$.

We have the following questions related to the problem (1):

- (i) Existence of the best approximation: is there an element $\mathbf{p}^* \in \mathcal{G}$ for which $\|\mathbf{f} - \mathbf{p}^*\| = E_{\mathcal{G}}(f)$?
- (ii) Unicity: if the property (1) defines uniquely the element $\mathbf{p}^*$?
- (iii) Besides (1), what else can we state with respect to $\mathbf{p}^*$?
- (iv) What are the *numerical* methods which can be used to find $\mathbf{p}^*$?

The case $\mathbf{f} \in \mathcal{G}$ gives a trivial problem: $\mathbf{p}^* = \mathbf{f}$ and $E_{\mathcal{G}}(f) = 0$.

Best approximation – existence

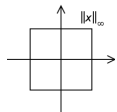
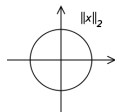
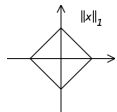
Theorem 11.1 *Given a normed vector space $\mathcal{V}$ and an element $\mathbf{f} \in \mathcal{V}$ there exists $\mathbf{p}^*$ best approximation in the n -dimensional subspace $\mathcal{G} \subseteq \mathcal{V}$.*

Proof. We do not give the proof of this. ■

Best approximation – unicity

Next, we study the unicity. We cannot give general results.

We take an example with the norms $\|\cdot\|_p, p = 1, 2, \infty$.



- So, we have seen that using norm $\|\cdot\|_2$ we can give exactly one solution.
- We need to introduce the concept of *strictly convex norms*.

Definition 11.2 *The linear space $\mathcal{V}$ has strictly convex norm, if given vectors $\mathbf{f}, \mathbf{h} \in \mathcal{V}$, which are both different from $\mathbf{0}$, we have*

$$\|\mathbf{f}\| = \|\mathbf{h}\| = 1 \Rightarrow \left\| \frac{\mathbf{f} + \mathbf{h}}{2} \right\| < 1.$$

We can use the following Lemma

Lemma 11.3 *The best approximations of $\mathbf{f}$ make a convex subset of the subspace $\mathcal{G}$.*

Proof. Let $\mathbf{p}^*$ and $\mathbf{p}^{**}$ be two best approximations of $\mathbf{f}$, and $\lambda \in [0, 1]$ arbitrary. Then

$$\begin{aligned}\|\mathbf{f} - (\lambda\mathbf{p}^* + (1 - \lambda)\mathbf{p}^{**})\| &= \|\lambda(\mathbf{f} - \mathbf{p}^*) + (1 - \lambda)(\mathbf{f} - \mathbf{p}^{**})\| \\ &\leq \lambda\|\mathbf{f} - \mathbf{p}^*\| + (1 - \lambda)\|\mathbf{f} - \mathbf{p}^{**}\| \\ &= \lambda E_{\mathcal{G}}(f) + (1 - \lambda)E_{\mathcal{G}}(f) \\ &= E_{\mathcal{G}}(f),\end{aligned}$$

so the convex combination of the two elements are also optimal. ■

Theorem 11.4 (*Best approximation – unicity*) If the vector space $\mathcal{V}$ has strictly convex norm, then for any $\mathbf{f} \in \mathcal{V}$ there exists at most one best approximation $\mathbf{p}^* \in \mathcal{G}$.

Proof. We give indirect proof. Let $\mathbf{p}^* \neq \mathbf{p}^{**}$ be two different best approximations. Due to Lemma 11.3 we have that $\mathbf{q}^* \stackrel{\text{def}}{=} \frac{1}{2}(\mathbf{p}^* + \mathbf{p}^{**})$ is also best approximation. We have that

$$E_{\mathcal{G}}(f) = \|\mathbf{f} - \mathbf{q}^*\| = \|\mathbf{f} - \mathbf{p}^*\| = \|\mathbf{f} - \mathbf{p}^{**}\|,$$

moreover

$$\|\mathbf{f} - \mathbf{q}^*\| = \left\| \mathbf{f} - \frac{1}{2}(\mathbf{p}^* + \mathbf{p}^{**}) \right\|$$

We know that the norm is strictly convex, hence

$$E_{\mathcal{G}}(f) = \left\| \mathbf{f} - \frac{1}{2}(\mathbf{p}^* + \mathbf{p}^{**}) \right\| = \left\| \frac{\mathbf{f} - \mathbf{p}^* + \mathbf{f} - \mathbf{p}^{**}}{2} \right\| < E_{\mathcal{G}}(f),$$

which is a contradiction.

Properties of best approximation

- The possible characteristics of the best approximations depend on the linear space $\mathcal{V}$ and norm $\|\cdot\|$
- In the followings we are considering the case of inner product, and the derived norm.
- Later, we will see the case of maximum norm.

Least squares approximation

Least squares approximation

- We call the linear space $\mathcal{V}$ as *inner product space*, if for two elements $\mathbf{f}, \mathbf{h}$ we have the inner product $\langle \mathbf{f}, \mathbf{h} \rangle$ defined, with the following properties: $\langle ., . \rangle : \mathcal{V} \times \mathcal{V} \rightarrow \mathcal{R}$ mapping:
 - (i) $\langle \mathbf{f}, \mathbf{f} \rangle \geq 0$ and if $\langle \mathbf{f}, \mathbf{f} \rangle = 0$, then $\mathbf{f} = \mathbf{0}$,
 - (ii) $\langle \mathbf{f}, \mathbf{h} \rangle = \langle \mathbf{h}, \mathbf{f} \rangle$,
 - (iii) $\langle \mathbf{f} + \mathbf{g}, \mathbf{h} \rangle = \langle \mathbf{f}, \mathbf{h} \rangle + \langle \mathbf{g}, \mathbf{h} \rangle$,
 - (iv) $\langle \alpha \mathbf{f}, \mathbf{h} \rangle = \alpha \langle \mathbf{f}, \mathbf{h} \rangle$.
- Two elements $\mathbf{f}$ and $\mathbf{h}$ of $\mathcal{V}$ are *orthogonal*, if $\langle \mathbf{f}, \mathbf{h} \rangle = 0$.

- From the inner product on $\mathcal{V}$ we can easily have a norm:

Theorem 11.5 *Given by $\|\mathbf{f}\| \stackrel{\text{def}}{=} \sqrt{\langle \mathbf{f}, \mathbf{f} \rangle}$ the mapping $\|\cdot\| : \mathcal{V} \rightarrow \mathcal{R}^+$ makes $\mathcal{V}$ into a normed vector space.*

- This is a very good news – the best approximation exists!

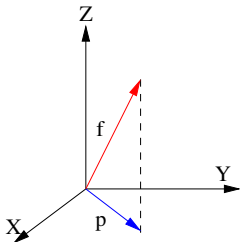
- In the followings, if we mention inner product spaces then we always refer to the normed vector space as it was defined on the previous slide.
- Similarly, if we refer the the subspace $\mathcal{G}$ of the inner product space $\mathcal{V}$, then $\mathcal{G}$ is also an inner product space.
- What are the properties of this norm?

Lemma 11.6 *All inner product spaces $\mathcal{V}$ are strictly convex normed.*

This is the second good news: there exists at most one best approximation.

Theorem 11.7 *Let $\mathcal{V}$ be a linear inner product space, $\mathcal{G}$ the subspace of $\mathcal{V}$ and $\mathbf{f} \in \mathcal{V}$. The element $\mathbf{p}^* \in \mathcal{G}$ is the best approximation of $\mathbf{f}$ if and only if*

$$\langle \mathbf{f} - \mathbf{p}^*, \mathbf{g} \rangle = 0 \quad \text{for any } \mathbf{g} \in \mathcal{G}. \quad (2)$$



- If $\mathcal{V}$ is a linear inner product space, and its subspace $\mathcal{G} = \text{span}(\mathbf{g}_1, \mathbf{g}_2, \dots, \mathbf{g}_n)$ is of finite dimension, then it follows from the previous result that the best approximation uniquely exists and it can be obtained by solving a *system of linear equations*.

Corollary 11.8 *The element $\mathbf{p}^* \in \mathcal{G}$ is the best approximation of $\mathbf{f}$ iff*

$$\langle \mathbf{f} - \mathbf{p}^*, \mathbf{g}_i \rangle = 0 \quad i = 1, 2, \dots, n. \quad (3)$$

If $\mathbf{p}^* = \sum_{k=1}^n \alpha_k \mathbf{g}_k$, then condition

$$\langle \mathbf{f} - \mathbf{p}^*, \mathbf{g}_i \rangle = 0 \quad (i = 1, 2, \dots, n) \quad (4)$$

given in the previous Corollary 11.8 is equivalent to this: the vector $(\alpha_1, \alpha_2, \dots, \alpha_n)^T$ is the solution of the *normal equations*

$$\sum_{k=1}^n \alpha_k \langle \mathbf{g}_k, \mathbf{g}_i \rangle = \langle \mathbf{f}, \mathbf{g}_i \rangle \quad (i = 1, 2, \dots, n) \quad (5)$$

The matrix

$$\mathbf{G} = \begin{pmatrix} \langle \mathbf{g}_1, \mathbf{g}_1 \rangle & \langle \mathbf{g}_2, \mathbf{g}_1 \rangle & \cdots & \langle \mathbf{g}_n, \mathbf{g}_1 \rangle \\ \langle \mathbf{g}_1, \mathbf{g}_2 \rangle & \langle \mathbf{g}_2, \mathbf{g}_2 \rangle & \cdots & \langle \mathbf{g}_n, \mathbf{g}_2 \rangle \\ \vdots & \vdots & \ddots & \vdots \\ \langle \mathbf{g}_1, \mathbf{g}_n \rangle & \langle \mathbf{g}_2, \mathbf{g}_n \rangle & \cdots & \langle \mathbf{g}_n, \mathbf{g}_n \rangle \end{pmatrix}$$

is called the *Gram-matrix* with the basis $\{\mathbf{g}_1, \mathbf{g}_2, \dots, \mathbf{g}_n\}$.

- Since the system can be uniquely solved we have $\det(\mathbf{G}) \neq 0$.
 - Following from the symmetry of inner product of real space $\mathcal{V}$ we get that matrix $\mathbf{G}$ is symmetric.
 - Moreover, it can be shown that $\mathbf{G}$ is positive definite.
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- We have even simpler form if we work with special basis vectors of $\mathcal{G}$: the basis $\{\mathbf{g}_1, \mathbf{g}_2, \dots, \mathbf{g}_n\}$ is *orthogonal basis*, if $\langle \mathbf{g}_i, \mathbf{g}_j \rangle = 0$ holds whenever $i \neq j$.
 - From the definition of inner product it follows that $\langle \mathbf{g}_i, \mathbf{g}_i \rangle \neq 0$.
 - If we additionally also have $\langle \mathbf{g}_i, \mathbf{g}_i \rangle = \|\mathbf{g}_i\|^2 = 1$, then we call $\{\mathbf{g}_1, \mathbf{g}_2, \dots, \mathbf{g}_n\}$ the *orthonormal basis* of the subspace $\mathcal{G}$.

- For a subspace $\mathcal{G}$ write up the normal equations using the orthonormal basis: we have then the identity matrix $\mathbf{I}$, hence we get the trivial system

$$\alpha_i = \langle \mathbf{f}, \mathbf{g}_i \rangle \quad i = 1, 2, \dots, n. \quad (6)$$

The solution of this is the constant vector on the right side.

- If the basis is orthogonal, then we have a system with diagonal matrix, and the solution can be obtained by n division:

$$\alpha_i = \frac{\langle \mathbf{f}, \mathbf{g}_i \rangle}{\langle \mathbf{g}_i, \mathbf{g}_i \rangle} \quad i = 1, 2, \dots, n. \quad (7)$$